

Introduction to Population Biology

Second Edition

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4

Exponential Growth

Darwin noted in Chapter 3 of *On the Origin of Species* that parents on average produce more offspring than are needed for replacement, and consequently populations can potentially increase geometrically. This is a key fact in his theory of natural selection, but why should there be overproduction of offspring? Consider the fate of populations that do not have this characteristic. Obviously, if, on average, individuals cannot fully replace themselves, the population will decline. Populations adopting an exact replacement strategy suffer the same fate, because there is always a chance that some individuals will die before they reproduce and so these populations will also decline as their reproductive base shrinks. Thus, although natural selection can select for any reproductive rate, providing that rate leaves the most descendants among competitors, only those populations where, on average, there is an overproduction of offspring will survive over the long term, the others being eliminated unless they are supplemented by immigration from other populations. However, for the species as a whole, overproduction is a necessity.

We can conclude that overproduction is one of the necessary conditions for the long-term survival of populations, allowing them to compensate for pre-reproductive losses and to recover from reductions in population size.

4.1 INTRODUCING DENSITY-INDEPENDENT GROWTH

In our mathematical models, population size is denoted by the symbol N and we use subscripts to indicate the size of the population at different times. So N_t is the size of the population at time t , N_1 and N_2 the sizes of the population at times 1 and 2, and so on. By convention we use N_0 (the size of the population at time 0) to indicate the starting time of population growth. The symbols Δ and δ denote changes in population size.

The units of time, t , will vary according to the type of organism we are studying. For rapidly growing populations like bacteria, t may be in minutes, whereas for many trees and vertebrates, t may be measured in years. We also use the symbols Δ and δ to denote intervals of time.

Most of us recognize that the series 1, 2, 4, 8, 16, 32, 64, and so on forms an exponential or geometric series. What is the difference in these two terms? Geometric growth is where the population is measured at fixed time intervals, whereas exponential growth is where the population is measured at any point in time. Thus, they amount to the same thing, the only difference being whether we measure time at discrete intervals or continuously.

4.2 DEVELOPING THE GEOMETRIC AND EXPONENTIAL GROWTH MODELS

We begin by considering three geometric series in Table 4.1, where the multiplication rate, λ , remains constant over each time step, Δt , for three populations (A, B and C). The change in population size over each time step ($\Delta N/\Delta t = N_{t+1} - N_t$) is calculated in Table 4.1 in order to derive some mathematical relationships between the various parameters.

An examination of Table 4.1 reveals that there is a constant ratio between the population size (N_t) at the start of each time interval and the change in population size over the time interval ($\Delta N/\Delta t$). The ratio is 1 for population A, 2 for population B and 3 for population C, and represents the growth rate, R_m , of each population. We can represent this general relationship by our first equation:

$$\frac{\Delta N}{\Delta t} = R_m N_t \quad (\text{Eqn 4.1})$$

where R_m equals the per capita (i.e. per individual) rate of increase over the time period Δt . A graphical representation of the results of Eqn 4.1 for the three populations is shown in Figure 4.1a.

Further examination of Table 4.1 shows that there is a relationship between the growth rate, R_m , and the multiplication rate, λ . In population A, $\lambda = 2$ and $R_m = 1$, in population B, $\lambda = 3$ and $R_m = 2$, and in population C, $\lambda = 4$ and $R_m = 3$. We can therefore represent this relationship by our next pair of equations:

$$R_m = \lambda - 1 \quad (\text{Eqn 4.2})$$

$$\lambda = R_m + 1 \quad (\text{Eqn 4.3})$$

Table 4.1 The relationships between population size (N), change in population size ($\Delta N/\Delta t$) and the population multiplication rate (λ) for three populations that are growing geometrically

Time (t)	Population A: $\lambda = 2$		Population B: $\lambda = 3$		Population C: $\lambda = 4$	
	N_t	$\Delta N/\Delta t$	N_t	$\Delta N/\Delta t$	N_t	$\Delta N/\Delta t$
0	1	$2 - 1 = 1$	1	$3 - 1 = 2$	1	$4 - 1 = 3$
1	2	$4 - 2 = 2$	3	$9 - 3 = 6$	4	$16 - 4 = 12$
2	4	$8 - 4 = 4$	9	$27 - 9 = 18$	16	$64 - 16 = 48$
3	8	$16 - 8 = 8$	27	$81 - 27 = 54$	64	$256 - 64 = 192$
4	16	16	81	162	256	768

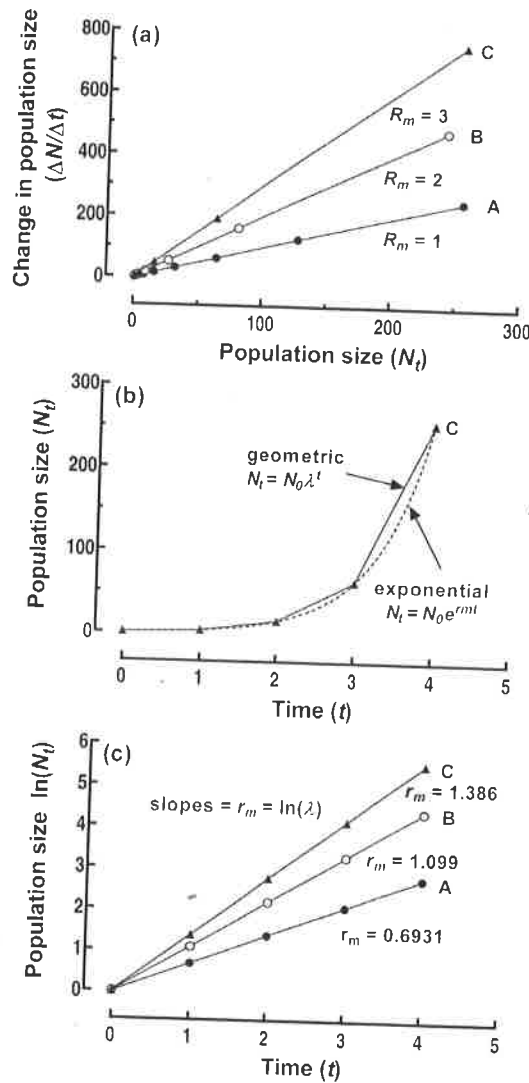


Figure 4.1 (a) The results of applying Eqn 4.1 to the three populations in Table 4.1.

(b) Comparing geometric growth (Eqn 4.5) and exponential growth (Eqn 4.7) when applied to population C in Table 4.1. (c) Showing that Eqns 4.10 and 4.11 provide identical solutions when applied to the three populations in Table 4.1.

If we simplify things by considering a closed population, where there is no immigration or emigration, the rate of increase (R_m) over the time period Δt will depend on the number of births (B) and deaths (D) during the time interval. However, to derive a per capita rate of increase, we need to divide the number of births and deaths by the total number at risk, i.e. the population size N at the start of the time interval. This gives us our fourth equation:

$$R_m = \frac{B - D}{N} \quad (\text{Eqn 4.4})$$

Finally, we can predict the population size after any number (t) of time steps as follows. The population size after one time step (N_1) will be the starting population time multiplied by the multiplication rate, i.e. $N_1 = N_0 \lambda$. Similarly, $N_2 = N_1 \lambda$, but $N_1 = N_0 \lambda$ and so substituting for N_1

we find that $N_2 = N_0\lambda\lambda$ or $N_2 = N_0\lambda^2$. We can do this for successive time steps to show that the general case is given by:

$$N_t = N_0\lambda^t \quad (\text{Eqn 4.5})$$

The form of population growth can now be predicted provided we know the starting number, N_0 , and the multiplication rate, λ , which is constant for each time interval (Figure 4.1b).

The equations we have just derived work perfectly provided the assumptions of the model are not violated. It also has a particular quirk in that the value of the multiplication rate, λ , is linked to the discrete time step, Δt . If we change the value of Δt , we cannot calculate the value of λ by simple scaling. For example, imagine that we determined that a bacterial population doubled in size every 20 min, i.e. $\lambda = 2$ and $\Delta t = 20$ min. If we wish to model the population at 10-min intervals, λ would not equal 1 (i.e. half of 2) because this would indicate that the population is not growing at all. Our next model of exponential growth provides a simple way of solving this quirk and so is more general.

If we make the time intervals Δt infinitesimally small, Eqn 4.1 becomes modified to the following differential equation:

$$\frac{\delta N}{\delta t} = r_m N \quad (\text{Eqn 4.6})$$

The expression $\delta N/\delta t$ represents the change in population size at an instant of time, and is the tangent to the population growth curve at population size N . The parameter r_m is the instantaneous rate of increase, sometimes called the intrinsic rate of natural increase or the Malthusian parameter after Thomas Malthus. The value of r_m is equal to the instantaneous birth rate (b) minus the instantaneous death rate (d) and so is analogous to R_m , except that the latter is measured over a discrete time step. We will show how to calculate r_m later.

To predict population size at any time (t), i.e. N_t , we integrate Eqn 4.6 following the rules of calculus. This is straightforward for anyone familiar with the rules of integral calculus (see Otto and Day 2007), but is unintelligible for those who are not. Do not worry if you don't know calculus. All we are doing by integrating an equation is to add up all of the small changes within defined limits. In this case, we add up the infinitesimally small changes in population size from time 0 (our starting time) to time t . The integral form is:

$$N_t = N_0 e^{r_m t} \quad (\text{Eqn 4.7})$$

This equation is the same as the formula for compound interest, where N_0 is the principle sum invested, r_m is the rate of interest and N_t is the accumulated total after time t . We can also note that Eqn 4.7 is similar in form to Eqn 4.5 (i.e. $N_t = N_0 \lambda^t$), and is illustrated in Figure 4.1b. A comparison of Eqns 4.5 and 4.7 reveals that:

$$e^{r_m} = \lambda \quad (\text{Eqn 4.8})$$

Taking the natural logarithm of both sides of this equation, and noting the relationships in Eqns 4.3 and 4.4, results in the following set of equivalent equations:

$$r_m = \ln(\lambda) = \ln(R_m + 1) = \ln\left(\frac{B - D}{N} + 1\right) \quad (\text{Eqn 4.9})$$

We can therefore calculate the value of r_m if we know either the population's multiplication rate, λ , or the birth and death rates over a finite time interval (Δt). Moreover, the value of r_m can easily be scaled from one set of time units to another. For example, if the value of r_m is 0.1 per day, then the r_m value per week is $0.1 \times 7 = 0.7$. In order to convert the multiplication rate, λ , from one time scale to another, however, one must first calculate the r_m value equivalent to λ , make the conversion, and then convert the new r_m value back to the new λ using Eqn 4.8. Thus, if λ per week is 2 and we wish to know the value of λ per day, we first calculate r_m per week using Eqn 4.9 ($= \ln(2) = 0.6931$), divide this by 7 to obtain r_m per day ($0.6931/7 = 0.0990$) and finally convert this value back to λ per day using Eqn 4.8 ($e^{0.099} = 1.104$). Ecologists tend to use r_m rather than λ , because it is very easy to compare the growth rates of different species that have been measured using different time scales. Population geneticists, however, use λ in their models because they are usually considering the growth of genotypes on a per-generation basis.

Finally, if we take the logarithm of both sides of Eqns 4.5 and 4.7, we obtain the following:

$$\ln(N_t) = \ln(N_0) + \ln(\lambda)t \quad (\text{Eqn 4.10})$$

$$\ln(N_t) = \ln(N_0) + r_mt \quad (\text{Eqn 4.11})$$

These two equations are equivalent. They show that the logarithm of population size changes linearly over time if the populations are growing exponentially or geometrically. Thus, we can see whether a population is growing exponentially by plotting the logarithm of population size over time. If it conforms to a straight line, the population is growing exponentially and the intrinsic rate of increase, r_m , is given by the slope of the graph (Figure 4.1c).

4.3 APPLYING THE GEOMETRIC AND EXPONENTIAL GROWTH MODELS

Many students will have found the last section on growth models to be rather unsatisfying. You may follow the logic of the algebraic proofs, but at the end it all seems so abstract. What does it all mean, what should one remember and how can one apply the model? To help answer these questions, we will now consider some examples of the use of the model.

EXAMPLE 4.1

Each individual of an annual plant species produces on average ten seeds. (a) Starting with a single individual, what will be the population size after 6 years, assuming all the seeds survive to produce adult plants the following generation? (b) A new variety of the plant produces 5 per cent more seeds, i.e. 10.5 seeds on average. What will be the potential number of offspring after 6 years from a single individual?

In (a), each individual produces ten seeds; therefore, the multiplication rate, λ , = 10 each year or generation. To find the potential population size (N_t) after 6 years (t), starting with a single individual (N_0), use Eqn 4.5 ($N_t = N_0\lambda^t$). Thus, $N_t = 1 \times 10^6 = 1\,000\,000$ offspring. (b) The new variety could potentially produce $N_t = 1 \times 10.5^6 = 1\,340\,096$ offspring over the same time period. Note that the new variety would have produced only 5 per cent more offspring in the first year but by the sixth year the difference has increased to approximately 34 per cent. If the two varieties are the same in their other characteristics, the new variety will steadily eliminate the old variety in their 'struggle for existence'.

EXAMPLE 4.2

The Norway rat breeds continuously and has an estimated rate of natural increase (r_m) of 0.0143 per day. A small number invade a garbage dump where living conditions are ideal. How long will it take the population to double in size?

First, rearrange Eqn 4.7 ($N_t = N_0e^{r_mt}$) to $N_t/N_0 = e^{r_mt}$. If the population doubles in size, $N_t/N_0 = 2$. Taking the logarithm of both sides of the rearranged equation, we have $\ln(2) = 0.6931 = 0.0143 \times t$, and so $t = 48.47$ or approximately 48 days.

EXAMPLE 4.3

A continuously growing population was observed to double in size every 3 days. Calculate the multiplication rate (λ) per day and per week.

The value of λ per 3 days = 2. First use Eqn 4.9 to calculate $r_m = \ln(2) = 0.6931$ per 3 days. Then calculate the r_m per day = $0.6931/3 = 0.2310$, and the r_m per week is $7 \times 0.2310 = 1.6173$. Then use Eqn 4.8 to calculate λ per day = $e^{0.2310} = 1.26$, and λ per week = $e^{1.6173} = 5.04$.

EXAMPLE 4.4

The Addo elephant population was observed to increase from 160 to 170 individuals from 1988 to 1989 as a result of 10 births and 0 deaths (Gough and Kerley 2006). What are the values of R_m , λ and r_m per year, and how long will it take the population to double in number assuming the growth rate remains constant?

To solve for R_m , we use Eqn 4.4, where $R_m = (10 - 0)/160 = 0.0625$. To solve for λ , we can use Eqn 4.3, where $\lambda = 0.0625 + 1 = 1.0625$; or we can rearrange Eqn 4.5 to $\lambda = N_1/N_0 = 170/160 = 1.0625$ (note $t = 1$). To solve for r_m , we use Eqn 4.9, where $r_m = \ln(1.0625) = 0.060625$. As expected, the value of r_m is slightly lower than the value of R_m (see Simulations website). To calculate the doubling time for the population, we can use either Eqn 4.5 or Eqn 4.7. If we use Eqn 4.5, we rearrange it to $N_t/N_0 = 2$ (doubling) = λ^t and take the logarithm of both sides to

obtain $\ln(2) = 0.6931 = 0.060625t$, and so $t = 11.43$ years. We would expect the population to double to 320 animals between 1999 and 2000. In fact there were 315 elephants in 1999 and 324 elephants in 2000. Not a bad estimate!

4.4 SIMULATING EXPONENTIAL AND GEOMETRIC GROWTH

Go to the Simulations website for Chapter 4 and follow the links to these simulations. I urge you to look at the simulations and follow the various exercises because you will better understand the various equations and see how they are related. Instructions are given to write your own programs in Excel, or you can simply use and modify the spreadsheets provided.

4.5 TWO EXAMPLES OF EXPONENTIAL GROWTH

In the city where I live (Saskatoon, Saskatchewan), there was a dramatic increase in the breeding population of the merlin falcon (*Falco columbarius*) throughout the 1970s and early 1980s (documented by Oliphant and Haug 1985). The arithmetic plot of the numbers of breeding pairs from 1970 to 1982 appears not to have had a smooth exponential growth form (Figure 4.2a), but a logarithmic plot of the numbers reveals that growth was approximately exponential during this time period (Figure 4.2b), i.e. the growth rate was approximately constant. The average intrinsic rate of natural increase, r_m , over the 13-year period was 0.236,

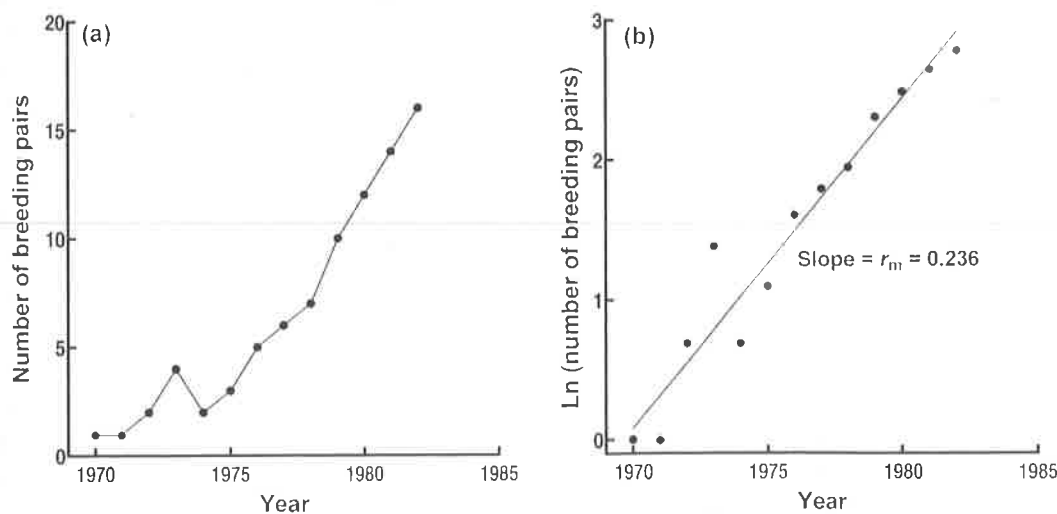


Figure 4.2 Growth of the breeding population of merlin falcons in Saskatoon from 1970 to 1982 on an arithmetic scale (a) and on a logarithmic scale (b). (Data from Oliphant and Haug 1985.)

giving an average yearly multiplication rate (e^r) of 1.266, or an average increase of almost 27 per cent per year.

Several factors combined to make this population increase possible. The prairie-parkland area of Canada has few suitable nesting habitats for merlins, but Saskatoon is located on the South Saskatchewan River where merlins occur naturally. The urban population was probably started from this source. Merlins do not build their own nests but take over old nests of American crow (*Corvus brachyrhynchos*) or black-billed magpie (*Pica pica*). In Saskatoon, these birds nest almost exclusively in large mature spruce trees that are older than 30 years. Thus, the seeds of the merlin invasion were set decades before 1970. Merlins thrive in the urban habitat because there is a large urban prey population of house sparrows (*Passer domesticus*). Once merlins had established themselves in the city, it is believed that the majority of new nests were established by birds fledged from city nests because the immigration of new birds from surrounding areas was limited. However, only the older parts of the city have mature spruce trees suitable for nesting. All of the nests found during the study period were located within a core area of 35 km² in the city, which in the early 1980s had a total area of 122 km². One can predict that as spruces begin to mature in newer neighbourhoods, and these areas are invaded by crows and magpies, the merlin population will expand into these areas.

There are many other species that have probably grown approximately exponentially when they have been introduced into new areas around the world and have been spectacularly successful in their new home. Examples include the prickly pear cactus (*Opuntia*) introduced into Australia, South Africa and the Hawaiian Islands from South America, the rabbit (*Oryctolagus*) introduced to Australia from England and the many species of sports fish that have been introduced to provide freshwater fishing throughout the world.

Our second example considers a population of elephants (*Loxodonta africana*) in the Addo area, located approximately 60 km north of Port Elizabeth, South Africa. Hunters had exterminated most of the elephants from the Eastern Cape by the early 1900s, and by 1931 there were only 11 elephants left, and the Addo Elephant National Park was established to protect them. The elephants were not totally protected, however, because the park was inadequately fenced, and those leaving the park were sometimes killed. Nevertheless, by 1954, the population had doubled to 22 elephants and the 22.7 km² area of the park was properly fenced to prevent any further emigration from the area. The population continued to grow more or less exponentially from 11 in 1931 to 388 in 2003, a period of 72 years, and there are no signs that the growth rate is declining (Figure 4.3). To help accommodate this increase, the park has been expanded five times since 1977 to the present fenced area of 103 km². The average yearly intrinsic rate of natural increase, r_m , was 0.05178 during this time, giving an average yearly multiplication rate (e^r) of 1.0531, or 5.3 per cent each year. Since 1976, the yearly multiplication rate has been 5.8 per cent (Gough and Kerley 2006). Starting in 2003, some male elephants have been captured and removed to other wildlife parks and other male elephants have been introduced. This is in response to the reduced genetic variability in this population. Unfortunately, it is difficult to obtain details about this development and the current size of the population.

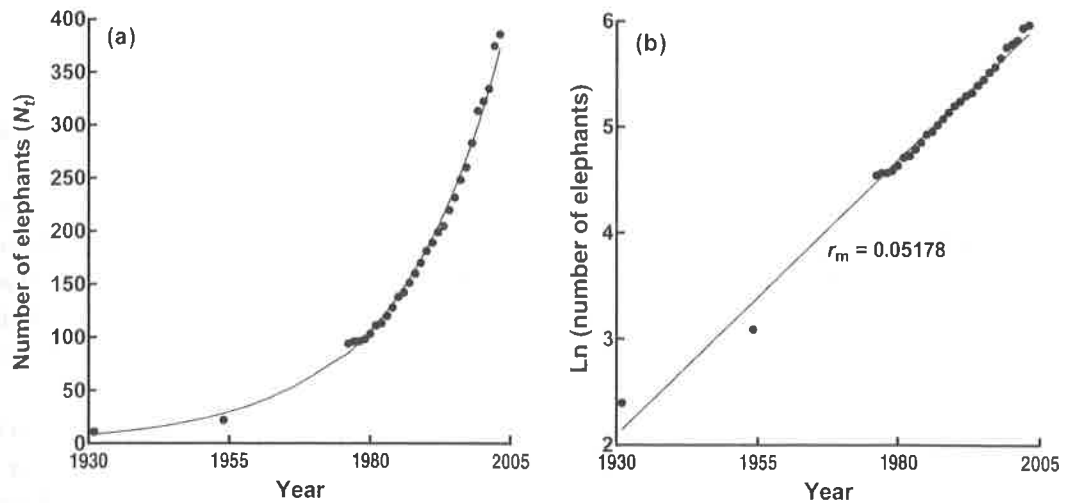


Figure 4.3 Growth of the Addo elephant population in South Africa from 1931 to 2003 on an arithmetic scale (a) and on a logarithmic scale (b). (Data from Gough and Kerley 2006, and earlier historical records.)

Similar rates of increase for elephant populations recovering from overhunting have been reported for Botswana and Zimbabwe (Cumming and Jones 2005). What about the future of the Addo Park? It is being expanded to form the Greater Addo National Park, an area of 3400 km², which is predicted to be able to support a sustainable population of approximately 2700 elephants (Woodd 1999), assuming 2 km² per elephant. Woodd (1999) predicted that the population size would reach 500 in the year 2010, 1000 in 2023, and 2700 in 2043. The corresponding estimates, assuming exponential growth using our data from Gough and Kerley (2006), are 500 elephants by 2008, 1000 elephants by 2021–2022, and 2700 elephants by 2040–2041. The estimates are reasonably compatible, and show that, unless methods are soon developed to reduce the rate of increase to zero, the elephant population will rapidly exceed the carrying capacity of the park.

We could provide numerous examples of exponential growth, but such unrestrained growth is unsustainable in the long term, and Darwin developed this point in Chapter 3 of *On the Origin of Species*. He noted that Linnaeus had calculated that a single annual plant producing only two seeds would have a million descendants after 20 years (actually $2^{20} = 1\,048\,576$). We know that some bacteria, such as *Escherichia coli*, can potentially divide (double) every 20 min and so have 72 generations during a single day. Consequently, a single individual can potentially increase to 4722 quintillion (i.e. 4722×10^{18}) individuals in 1 day, and we should remember that half of this total would have been produced during the last 20 min! Darwin also calculated that a pair of elephants, which breed very slowly, could produce 15 million descendants after 500 years. His estimate was overly cautious. If we use the information from our second example of exponential growth, where $r_m = 0.05178$ per year, a single pair of elephants would produce 175 341 265 415 descendants after 500 years! We can conclude that such growth is unsustainable.

4.6 Summary and Conclusions

The long-term survival of species requires them to produce more offspring than are needed to replace themselves in order to compensate for pre-reproductive mortality and also to recover from any reductions in population size. Consequently, populations of species can increase in size, and if the rate of increase is constant over a number of generations, the population growth is exponential or geometric. Both of these mathematical models have been developed and used to analyse field data and to answer various problems about population growth; for example, it was shown that the doubling time of population growth remains constant and that the logarithm of population size increases linearly in both of these models.

However, exponential increases in population size are unsustainable in the long term because the resources that are required for growth are finite. Such growth can be sustained for surprisingly long periods, as shown by the Addo elephant population for over 70 years and the human population for hundreds of years, but we can predict with certainty that such growth is unsustainable.

4.7 Problems

To check your understanding of this chapter, try the following problems. A summary of equations is provided in Box 4.1, and the answers to the problems may be found in the Solutions section at the end of the book.

1. A moth has an annual life cycle. One population was observed to increase from 5000 to 6000 individuals in 1 year. Predict the size of the population after 3 years (from the starting population of 5000), assuming no change in the rate of growth.
2. The human population increased from approximately 600 million to 900 million between 1700 and 1800 AD. Calculate the value of r_m and λ per year, assuming exponential growth.
3. A small population of kudu, introduced into a wildlife reserve area, is observed to increase by 5 per cent on average every year. Approximately how many years will it take for the population to double in size?
4. The value of r_m for a rat population is 0.14 per week. Starting with a population of 24 rats, what will be the approximate population size after 65 days, assuming exponential growth?
5. A population increases fivefold over a 4-week period. What is the value of λ per day, assuming exponential growth?
6. It was estimated that in 1959 the world's human population was 2 907 000 000 with an overall birth rate of 36 per 1000 people per year and a death rate of 19 per 1000 people per year. What was the expected increase in population size in 1959?

7. In 1959, the human population of the world increased by approximately 50 million from 2 907 000 000. (a) Calculate the value of r_m per year to four decimal places, assuming exponential growth; and (b) if the average death rate was 19 per 1000 people per year, what was the average birth rate in 1959?

BOX 4.1 SUMMARY OF EQUATIONS

Geometric growth model

$$\frac{\Delta N}{\Delta t} = R_m N_t \quad (\text{Eqn 4.1})$$

$$R_m = \lambda - 1 \quad (\text{Eqn 4.2})$$

$$\lambda = R_m + 1 \quad (\text{Eqn 4.3})$$

$$R_m = \frac{B - D}{N} \quad (\text{Eqn 4.4})$$

$$N_t = N_0 \lambda^t \quad (\text{Eqn 4.5})$$

$$\ln(N_t) = \ln(N_0) + \ln(\lambda)t \quad (\text{Eqn 4.10})$$

Exponential growth model

$$\frac{\delta N}{\delta t} = r_m N \quad (\text{Eqn 4.6})$$

$$N_t = N_0 e^{r_m t} \quad (\text{Eqn 4.7})$$

$$e^{r_m} = \lambda \quad (\text{Eqn 4.8})$$

$$r_m = \ln(\lambda) \quad (\text{Eqn 4.9})$$

$$\ln(N_t) = \ln(N_0) + r_m t \quad (\text{Eqn 4.11})$$

Note that the per capita (i.e. individual) rate of increase (R_m) is measured over the entire duration of the time step Δt , whereas the intrinsic rate of increase (r_m) is measured over an infinitesimally small time step. In both cases, they equal the per capita births less the per capita deaths (ignoring immigration and emigration), but they relate to different time frames. In some texts, the multiplication rate (λ) is given the symbol R , so be careful when you compare equations from different sources.